

SIEMENS

MAGNETOM

MR

Maintenance Instructions

System

Trio, A Tim System

The protocol M6-030.832.01.01.02 is required for these instructions

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1 General information

1.1 General

NOTE

These maintenance instructions comply with EC guideline CE 93/42/EWG.

 WARNING

Please follow the General Safety Notes as well as the MR-Specific Safety Information.

Non-compliance with safety notes may result in death or serious injury.

- ⇒ Carefully read the safety information in the General Safety Notes as well as the MR-Specific Safety Information before starting work activities. Safety warnings have to be observed at all times.
-

1.2 General Remarks

To ensure the continuous safety, quality, and reliability of these products, periodic maintenance has to be performed in accordance with the instructions in this manual as well as those in the “Safety-related tests”.

Periodic maintenance includes:

- Safety-related tests (not included in these maintenance instructions)
- Preventive maintenance
- Software maintenance
- Image quality assurance

Image quality assurance should always be performed upon completion of maintenance activities.

1.3 Documentation

A protocol of the maintenance activity has to be generated. All fields have to be completed. If a field does not apply to the system or if there is no information to be entered, enter 'n.a.' in the field.

Each work step or test performed successfully has to be confirmed in the test protocol by placing a checkmark next to "O.K." Work steps or tests that cannot be performed (e.g. options) or which are not performed (e.g. interval is more than one year) should be marked with "n.a." If an error occurs that cannot be eliminated, mark the respective test with "not O.K."

The maintenance protocol has to be archived by the operator in the System Owner Manual. We recommend that the country organizations also compile such documentation to ensure optimal operation of products, e.g. in the event damages occur.

1.4 Explanation of the individual steps

By explaining the different steps or maintenance modules, we hope to help you divide the maintenance procedure into workable units. At the same time, our objective is to describe the actions covered by the various modules.

We recommend that you demonstrate all processes to the customer. The more informed our customers are, the more they'll appreciate our efforts to provide optimal service.

Thank you for your cooperation.

1.4.1 Safety-related tests

Safety-related tests have to be performed according to the instructions in "Safety-related tests" and documented using the "Safety-related tests protocol".

1.4.2 Preventive Maintenance

The inspection of moving parts includes checking and replacing all wear and tear parts (e.g. filters, blowers, batteries, control lamps, etc.).

It also includes a visual inspection of the component and adjacent magnet for signs of wear or defective components. The temperature of the MR suite should be checked, as well as the collision-free zones required to perform updates in a safe environment (e.g. no boxes or other objects are blocking exits or air inlets, etc.).

1.4.3 Software

To maintain high software availability, a description of how to save dynamic data is included in this chapter.

1.4.4 Image quality assurance

To provide the customer, if requested, with performance data (regarding image quality) image quality assurance tests may be performed on a regular basis. The results are entered in a protocol filed at the customer site.

The quality measurements have to be performed for safety reasons and to verify the functionality of the overall system.

1.5 Acronyms:

Abbr.	Explanation
SI	Safety Inspection
SIE	Safety Inspection Electrical Safety
SIM	Safety Inspection Mechanical Safety
PM	Preventive Maintenance
PMP	Preventive Maintenance Preventive Parts Exchange, External Inspection, etc.
PMA	Preventive Maintenance Adjustments
PMF	Preventive Maintenance Function, Operating Value Check
Q	Quality Check
QIQ	Quality Check Image
QSQ	Quality Check System
SW	Software Maintenance

1.6 Prerequisites

1.6.1 Service tools or equipment needed for Preventive Maintenance

Tab. 1 Preventive Maintenance

Quantity	Name	Part number
1	Adsorber (every two years) - APD type	74 61 127
1	Wrench 37 mm (commercially available)	n.a.
1	Spotmeter SMFit Act	77 52 848
1	Set of standard tools	n.a.
1	Bucket 5 l - 10 l	n.a.
1	Vacuum pump filter (for optional comfort kit only; every two years)	55 16 930
2	Batteries (for optional patient trolley only; type Alkaline-Manganese "AA")	n.a.

1.6.2 Service tools or equipment needed for Software Maintenance:

Tab. 2 Software Maintenance

Quantity	Name
1	CD Recordable Note: Use only CD-R 's with medical grade e.g. Siemens Medical CD-R 80 or TDK Medical CD-R.

1.6.3 Service tools or equipment needed for QA

Tab. 3 QA

Quantity	Name
1	Spherical phantom, large; with holder
1	Loaded body phantom

1.7 Time frame for the maintenance schedule

Activity	Interval (every)	Time required	Note
Preventive Maintenance	12 months	150 min.	
- Replacing the adsorber	24 months	60 min.	
Software Maintenance	12 months	30 min.	
Image Quality Assurance	12 months	70 - 120 min. ¹	recommended every 3 months

1. Depending on installed options

Activity	Interval	Time required	Note
Replacing the displacer	¹	4 hours	if defective, see description in 'Replacement of Parts'
Refilling helium (e.g. 2 x 450 l)	²	5 hours	Refer to 'Start-up'

1. Interval is variable

2. Interval is variable

2 Preventive Maintenance

2.1 Cooling

Required time: 10 min.

Required tools: n.a.

PMP General checks

Check the entire water system for leaks and condensation, especially all water connections and water hoses within the:

- ACC
- GPA
- ACS
- SEP or IFP (option)

2.2 GPA, ACC Cabinet

Required time: 10 min.

Required tools: Screwdriver

PMP Checking the blower assembly

The blower assembly is integrated in the ACC cabinet.

- Switch off the GPA and ACC cabinets.
- Carefully swivel out the electronic section of the ACC.



WARNING

High Voltage!

Contact with live parts may result in death or serious injury.

⇒ **Keep away from live parts**

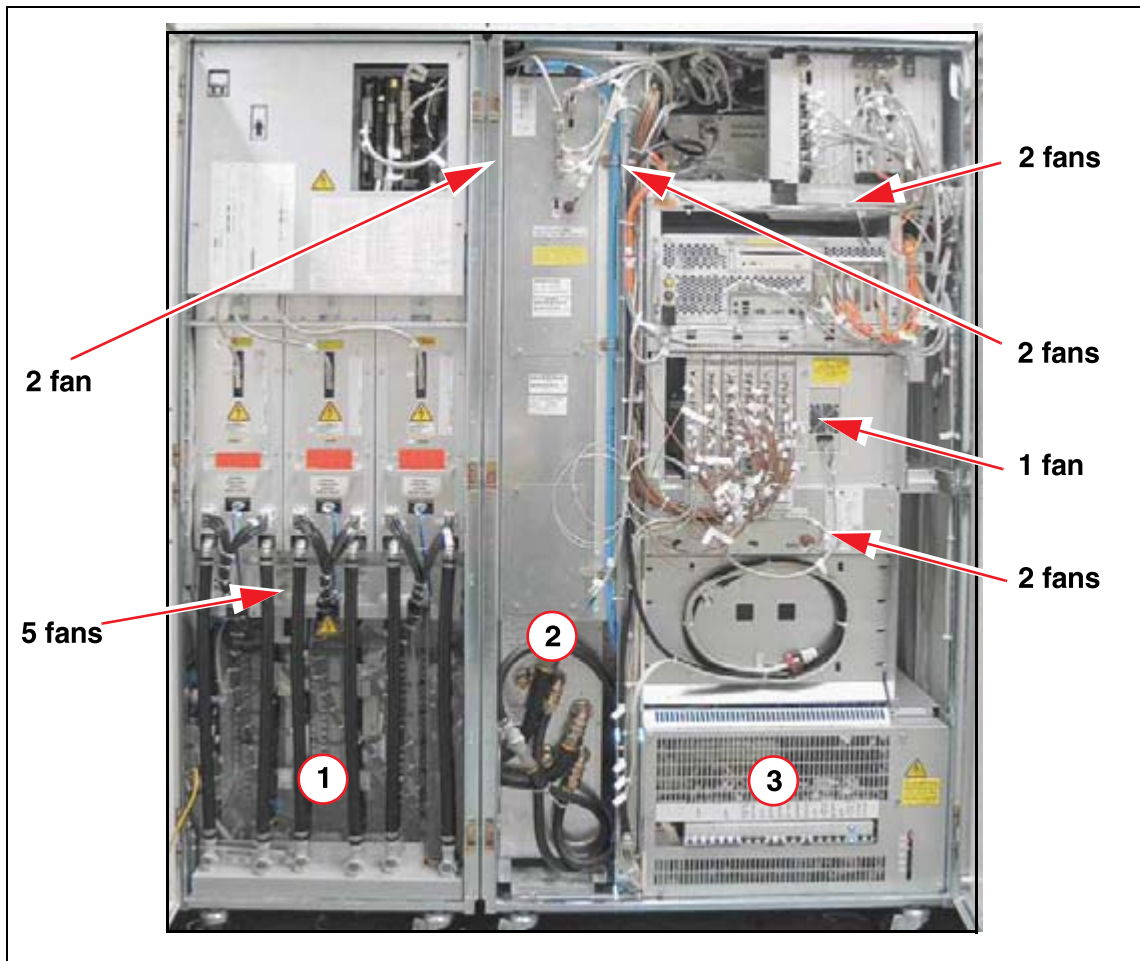


Fig. 1: GPA - ACS - ACC

Pos. 1 GPA

Pos. 2 ACS

Pos. 3 ACC

- Switch on the GPA and ACC cabinets.

- Visually inspect the operation of the fans and blowers in the ACC cabinet by using a flashlight. Replace, if necessary.

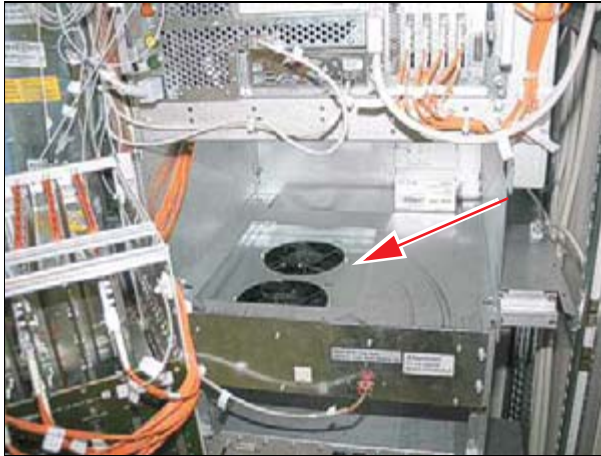


Fig. 2: Fans in ACC view

CAUTION

Fans in operation

Sharp edges may cause injury.

⇒ **Do not touch blowers and fans while in operation.**

- Switch off the GPA and ACC cabinets.
- Carefully swivel in the electronic section of the ACC.

2.3 RFPA 35KW

PMP Cleaning / replacing the RFPA air filter

Required time: 10 min.

Required tools: n.a.

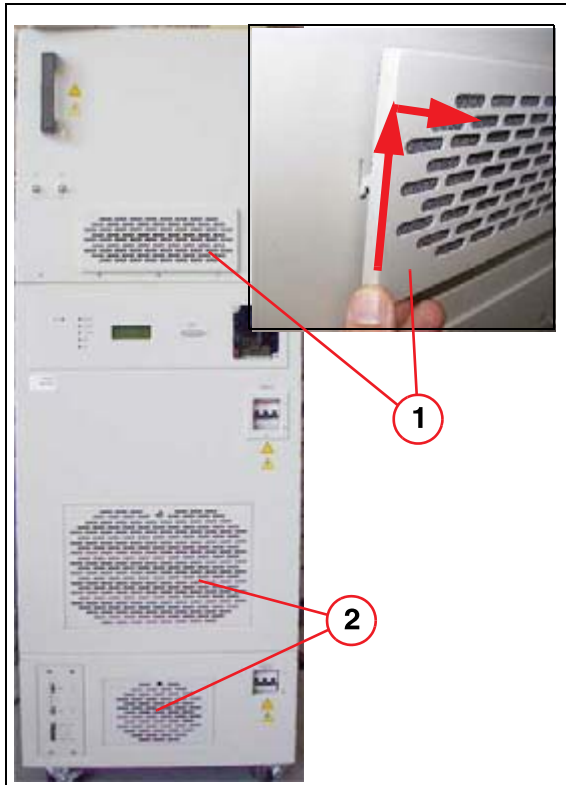


Fig. 3: RFPA 35 KW

- Slide the filter frame (1) upward and remove the filter .
- Check the filter pad for contamination and damage, and clean / replace, if necessary.
- Reassemble the air filter.
- Loosen the screws and open the filter frame (2).
- Check the filter pads for contamination and damage, and clean / replace if necessary.
- Reassemble the air filters.

2.4 SEP (option)

Required time: 20 min.

Required tools: 10 mm wrench, bucket, water hose for filling, 37 mm wrench

Abbreviation: clockwise - cw; counterclockwise - ccw

- Switch off F102 on the top of the ACC cabinet.
- Acknowledge the alarm on the alarm box.



Fig. 4: SEP roof

Pos. 1 Shut-off valves (primary water circuit)

Close the shut-off valves (1) to isolate the primary water circuit.

- Turn the shut-off valves ccw by 90° using a 10 mm wrench.

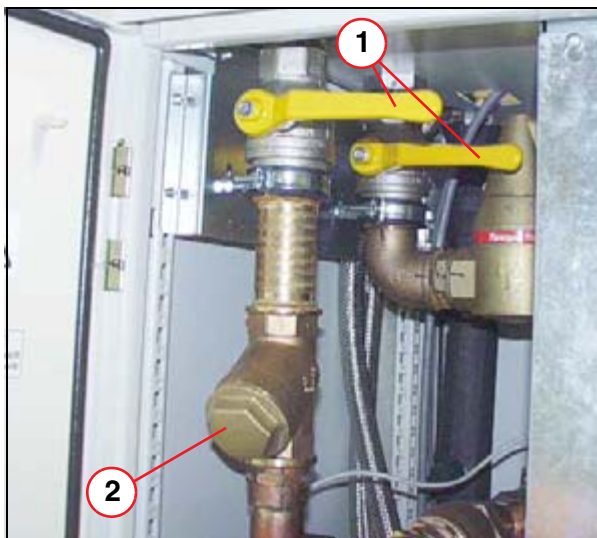


Fig. 5: Shut-off valve/Filter (secondary water circuit)

Close the shut-off valves (1) to isolate the secondary water circuit.

- Turn the shut-off valves ccw 90°.

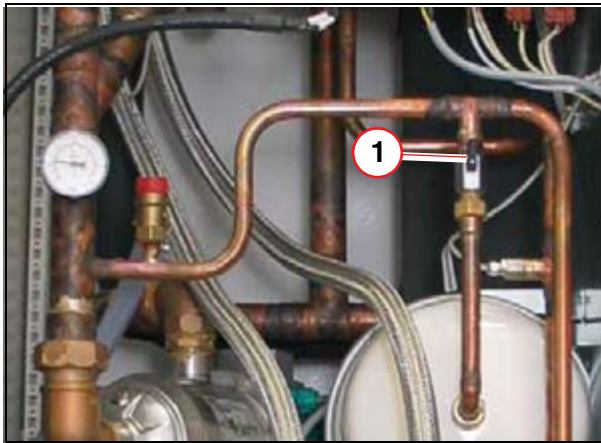


Fig. 6: Expansion chamber
Pos. 1 Valve

- Close the valve at the expansion chamber.

PMP Cleaning the strainer of the primary water circuit

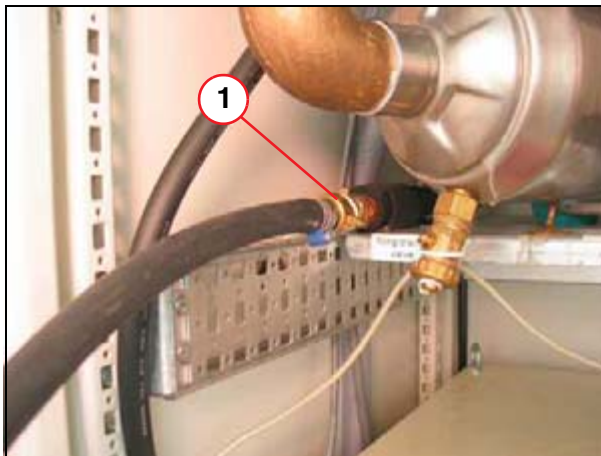


Fig. 7: Fill and drain valve

- Connect the water hose to the primary filling and drain port (Fig. 7 / p. 16) and run the hose to a drain container.
- Open the drain port and drain. The volume of water to be drained from the water circuit approx. 3 liters.
- Carefully remove the strainer cap using a 37 mm wrench to speed up the draining process ((1/ Fig. 8 / p. 17) primary water circuit).
- Close the drain port.



Fig. 8: SEP roof
Pos. 1 Strainer (primary water circuit)

- Open the strainer by using a 37 mm wrench ((1/ Fig. 8 / p. 17) primary water circuit).
- Inspect the strainer for contamination and clean it by rinsing under tap water.
- Install the strainer.

PMP Cleaning the strainer of the secondary water circuit



Fig. 9: Fill and drain valve

- Connect the water hose to the secondary filling- and drain port and run the hose to a drain container.

The volume of water to be drained from the water circuit is approx. 3 liters.

- Open the drain port and drain.
- Carefully remove the strainer cap using a 37 mm wrench to speed up the draining process (secondary water circuit).
- Close the drain port.

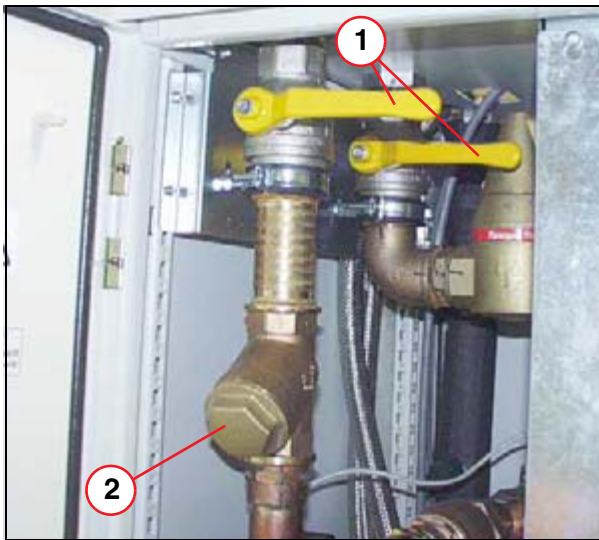


Fig. 10: Shut-off valve/Filter (secondary water circuit)

- Open the strainer (2) by using a 37 mm wrench (secondary water circuit). Inspect the strainer for contamination and clean it by rinsing under tap water.
- Install the strainer.



Fig. 11: Water filling

- Connect water hose from the water supply to the refilling port of the pump ([Fig. 9 / p. 17](#)).



Fig. 12: SEP manometer

- Open the drain valve.
- Open the shut-off valves for the primary and secondary water circuit.
- Open the valve of the expansion chamber.
- Refill the SEP with water until the filling pressure is reached (1/ Fig. 12 / p. 19).

There are two different filling pressures depending on the SEP version (Fig. 13 / p. 19), (Fig. 14 / p. 19).



Fig. 13: SEP old version (with pressure switch)
Pos. 1 Pressure switch



Fig. 14: SEP new version (without pressure switch)

- ⇒ **Old version** : The filling pressure is approximately 1.5 - 2.0 bar (Fig. 13 / p. 19).
- ⇒ **New version** : The filling pressure is approximately 0.5 - 1.0 bar (Fig. 14 / p. 19).
- Switch on F102 on the top of the ACC cabinet.
- Close the drain valve and disconnect water hose.
 - ⇒ The cold head is running.

2.5 IFP (option)

Required time: 20 min.

Required tools: 37 mm wrench

PMP Cleaning of the strainer

1. Switch off the helium compressor.



Fig. 15: Chiller

Pos. 1 ON - OFF Switch

- Switch off the chiller (1).

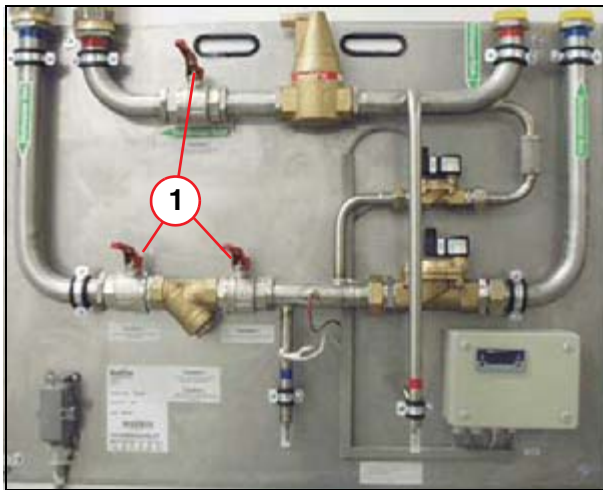


Fig. 16: IFP shut-off valves

- Slowly close the shut-off valves.



Fig. 17: IFP filter

1. Open the strainer (1) by using a 37 mm wrench.
2. Inspect the strainer for contamination and clean it by rinsing under tap water.
3. Install the strainer.
1. Slowly open the shut-off valves.

1. Switch on the chiller.
2. Switch on the helium compressor.

2.6 Magnet system

2.6.1 Center frequency

Required time: 5 min.

Required tools: n.a.

PMP Measuring the center frequency

- Register patient.
- Select protocol.
- Select <Options>, <Adjustments> in the main menu.
 - ⇒ The current center frequency will be displayed under Frequency (sys.).

Tolerance range for center frequency: 123.100 MHz - 123.265 MHz.

The magnet's field decay is: approx. 6.25 KHz / month

NOTE

Prior to reaching the lower limit of the tolerance range, the magnet has to be scheduled for re-ramping during the next maintenance.

For the ramping procedure, refer to "Technical Documents," register Start-up.

2.6.2 Ice formation and condensation

PMP Checking the service turret for external ice formation and condensation

Required time: 10 min.

Required tools: n.a.

- Remove service access cover
 - ⇒ In general, the service turret should be free of massive condensation.
- Empty out the water from the condensation reservoir located at the electronics side of the magnet.

2.6.3 Shield temperature

Required time: 5 min.

Required tools: n.a.

PMP Checking the shield temperature of the magnet

- Check the shield temperature using ' **Magnet & Cooling > (Magnet) Status** ' in the Service Software.
 - ⇒ In case of shield temperatures > 75 K, cooling capacity is too low. The displacer on the cold head needs to be replaced.

2.6.4 Adsorber replacement

Required time: 60 min.

PM Replacing the adsorber (every two years)

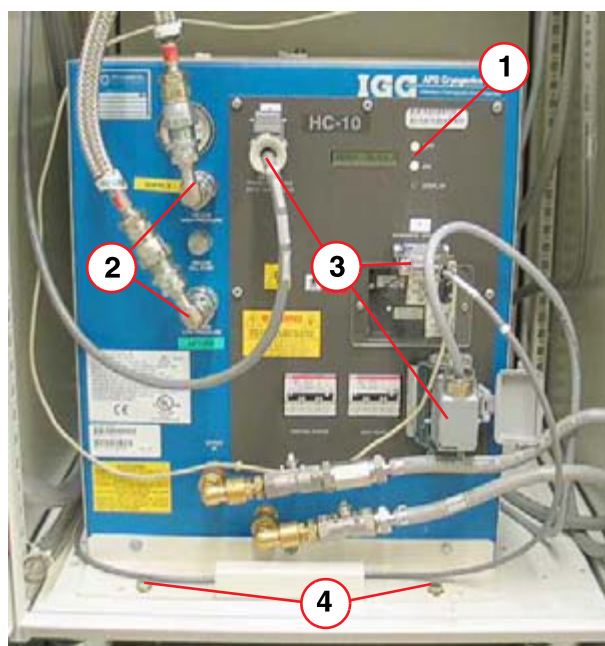
The adsorber has to be replaced every two years.

The compressor is stand-alone or mounted in the SEP / RCA.

2.6.4.1 Materials and tools required

- Adsorber
- Medium Philips screwdriver, large Philips screwdriver
- Set of nuts
- Fork wrench: 13mm
- Fork wrench: 1 1/8"; 1 3/8"; included with magnet accessories

2.6.4.2 Compressor removal



- Switch off compressor(1).
- Switch off MAIN POWER and CONTROL SYSTEM circuit breaker
- Unplug all cables (3).
- Remove screws (4).

Fig. 18: APD compressor

- Pos. 1 ON / OFF switches
Pos. 2 Helium flexlines
Pos. 3 Cables
Pos. 4 Attachment screws

2.6.4.3 Adsorber exchange

NOTE

Always hold the stationary nut on the gas line coupling with one wrench while turning the moveable coupling with the other wrench.

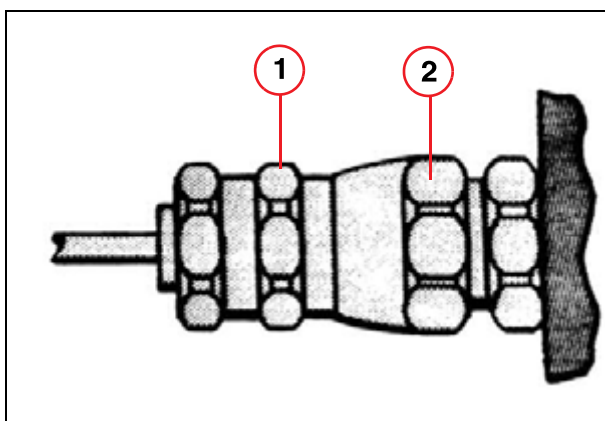


Fig. 19: Helium flexlines coupling

Pos. 1 Nut 1 1/8"

Pos. 2 Nut 1 3/16"

Remove helium flexlines

- Hold nut (1) using fork wrench (1 1/8") and loosen nut (2) using fork wrench (1 3/16").

There is no need to remove the water lines.

NOTICE

Avoid equipment failure

- ⇒ Do not tip the compressor by more than 10 degrees from horizontal, to avoid oil from flowing into unwanted places.

NOTE

The weight of the compressor is approximately 100 kg.

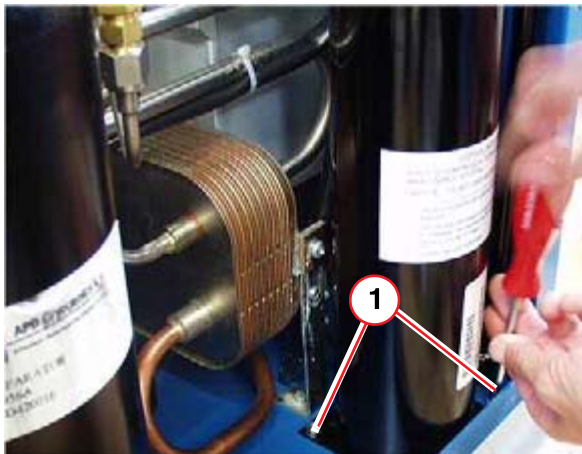


Fig. 20: Compressor out

- Carefully pull out the compressor.

The compressor should be pulled onto a sturdy support (e.g., wood) of approx. 12 cm.

- Remove housing screws. Lift housing cover at front end, pull forward and remove.



Remove Philips screws and adsorber .

Fig. 21: Adsorber support



Remove nut.

Fig. 22: Compressor, front panel

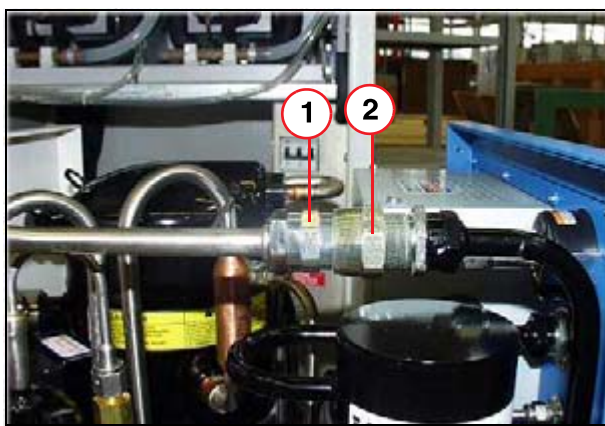


Fig. 23: Adsorber coupling

Pos. 1 Nut $1\frac{1}{8}$ "
Pos. 2 Nut $1\frac{3}{16}$ "

Unscrew helium gas coupling.

- Hold nut (1) using fork wrench ($1\frac{1}{8}$ ") and loosen nut (2) using fork wrench ($1\frac{3}{16}$ ") .



Fig. 24: Adsorber

- Remove adsorber.
- Retain all hardware to reuse with the new adsorber.

NOTE

The adsorber is filled with 99.99 grade helium at a pressure of 14-16 bar.

Do not vent the new adsorber

- Remove the dummy caps from the new adsorber.
- Install the lock washer onto the supply coupling of the new adsorber. Insert the supply coupling through the front panel and position the adsorber
- Insert and tighten the screws to fasten the new adsorber to the base.
- Install the nylon washer and the locknut on the supply coupling.
- Connect the adsorber's self-sealing coupling on its inlet side to the oil separator's outlet coupling.
- Assemble compressor housing.
- Secure compressor in the cabinet.
- Attach helium flex lines .
 - High pressure (red)
 - Low pressure (green)

2.6.4.4 Switch-on procedure

1. Close MAIN POWER and CONTROL SYSTEM circuit breaker on the compressor.

2. Switch on the compressor.

2.6.4.5 Safe disposal of the used adsorber



The adsorber is charged with helium gas!

Improper handling of the used adsorber may cause injury.

- ⇒ **Follow the venting procedure for safe disposal of the used adsorber.**

A venting adapter fitting is included with the new adsorber.

- Attach it to one of the self-sealing couplings on the used adsorber.
- Vent the used adsorber to atmospheric pressure.
- Turn the used adsorber over to a disposal company

2.6.5 Refrigerator system

Required time: 10 min.

Required tools: n.a.

PMP Checking the pressure values of the refrigerator system

2.6.5.1 APD refrigerator

- ⇒ Typical operating pressure: 20.0 - 22.8 bar (290 - 330 psig) at 50 Hz or 60 Hz.
- ⇒ With the compressor switched off, the fill pressure should be: 14.9 - 15.2 bar (216 - 220 psig) at 50 Hz or 60 Hz .

- Check for leaks if helium needed to be refilled (refer to TSG).

2.6.5.2 Helium compressor

Required time: 10 min.

Required parts and tools: n.a.

PMP Checking for leaks on the helium compressor

- Check for oil leaks on the compressor.
- Check for water leaks.
 - ⇒ Note the areas where condensation collects. The water droplets may be confused with leaks.

2.7 TFT Monitor

Required time: 15 min.

Required tools: Spotmeter SMFit Act

PMA Checking the TFT monitor adjustments

- Check the MRC TFT monitor adjustments using the SMFit Act spotmeter and test pattern.
- Check the MRSC TFT monitor (optional) following the same procedure.
- Check the In-Room MRC TFT monitor (optional) following the same procedure.

The adjustment procedure is described in 'Start-up'.

In case of deviations, the adjustment directions are contained in the documentation 'Replacement of Parts'

Checking the patient TFT monitor is unnecessary.

2.8 Phantoms

Required time: 15 min.

Required tools: n.a.

PMP Checking the phantoms for damage

Please read the “MR-specific safety information” prior to inspecting the phantoms.

Inspect every phantom used on-site for cracks, leaks, or extensive wear and tear.

Replace the phantom if there is any evidence of damage that may affect normal operation.

NOTE

Defective phantoms must be disposed of. Refer to the “Disposal Instructions”.

PMP Checking the phantoms for air bubbles

Inspect every phantom used on-site for air bubbles. Air bubbles can influence your tune-up results and quality measurements.

Ensure that fluid is at an adequate level so that measurement performance is not adversely affected.

It is normal for phantom fluid to evaporate over a period of time.

- Refill this phantom, if necessary.

NOTE

Phantoms with distilled water mixture - fill with distilled water only. Oil phantoms fill with the corresponding oil only.

2.9 RF Room Door

Required time: 15 min.

Required tools: 10 cm wide strip of paper,

PMP Checking the RF room door springs

- Check springs for mechanical damage. The springs should bounce back. Test the line contact at the outer surface of the contact by inserting a 10 cm wide strip of paper and pulling it out. Repair minor damage using appropriate tools, e.g. flatnose pliers.
- The metal of the contact surfaces and contact springs has to be clean. Remove any dirt, contamination, or hardened grease.

PMP Checking the RF room door hinges

- Check the hinges for cracks or possible damage.

PM Checking the RF room door lock

- The locking and unlocking mechanism has to function properly.
- The door handle/knob has to be securely fastened.

NOTE

In case of a defect (e.g. defective unlocking mechanism, door handle), inform the customer to initiate corrective action immediately.

2.10 Patient Trolley (option)

Required time: 10 min.

Required tools: 2 batteries (type “AA”)

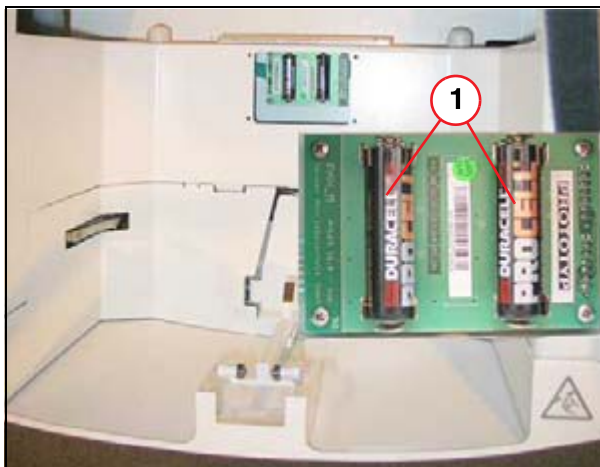
PM Replacing the patient trolley battery (option)



Remove the battery cover.

Fig. 25: Patient trolley

Pos. 1 Battery cover



- Replace the batteries (battery type “AA”).
- Mount the battery cover.
- Turn the old batteries over to a disposal company in accordance with local regulations.

Fig. 26: Patient trolley

Pos. 1 2 Mignon batteries

2.11 Vacuum pump (Comfort kit option)

Required time: 15 min.

Required tools: Filter (part no. 5516930 KE 625), Phillips screwdriver,

PMP Replacing the vacuum pump filter (every two years)

Location: filter plate

Over time the filter gets clogged with dust particles. For this reason, it is necessary to replace it frequently.



Filter (part no. 5516930 KE 625)

Fig. 27: Vacuum pump filter

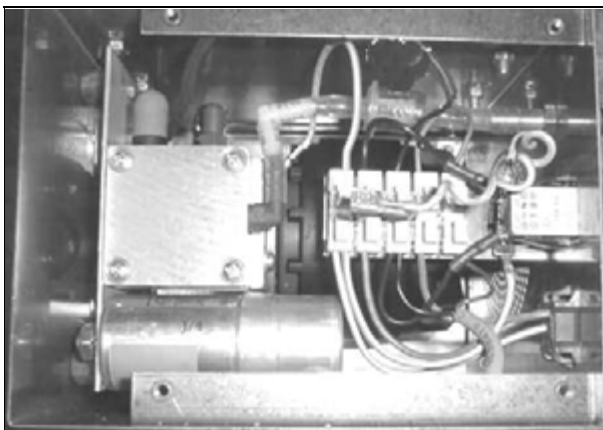


Fig. 28: Vacuum pump

- Switch off the system.
- Disconnect the power plug.
- Open the pump compartment (four Phillips screws).
 - ⇒ The vacuum connector is on the right.

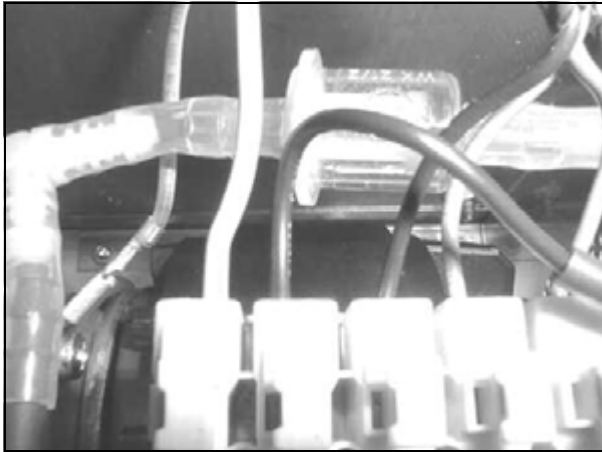


Fig. 29: Vacuum pump filter

- Replace the old filter with the new filter by simply pulling off the transparent hoses and refitting them to the new filter.

⇒ The filter is fitted in reverse direction.

(Observe arrow on filter.)

- Switch on the system.
- Connect the power plug.

3 Software maintenance

3.1 Saving site-specific data

Required time: 20 min.

Required tools / parts: CD-R

NOTE

Use only CD-R 's with medical grade, e.g. Siemens Medical CD-R 80 or TDK Medical CD-R.

- ⇒ Saving site-specific data should be performed once a year, but we recommend to perform it every three months.

When you are required to reinstall software (e.g. because of a defective system disk), the set of dynamic data will help you put the system back in operation as quickly as possible.

For this reason, we recommend to save the site-specific data each time you make changes in the site-specific configuration, tune-up, or sequences and protocols, even if the three-month interval has not expired.

NOTE

Backup is only supported as a local service for removable devices.

NOTE

Only raw (empty) CD-R media may be used to save site-specific data.

Settings that are not part of the backup/restore

The following settings are not part of the site-specific data. Make sure you keep the settings in your configuration pages.

- **DB offset**
- **Printer installation, printer configuration**

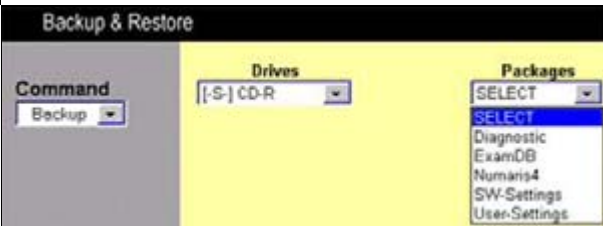
NOTE

In case the error message "C:\MedCom\MriSitedata\ContextVision\siemensIRU-PCNT-init-> access denied" is displayed during restore of data, re-enter the cv-license as described in 'Installation Software'.

SW Saving site-specific data

- Start the service platform
- Select **Backup & Restore**

Tab. 1

Output	Input	Note
 Fig. 1:	Select Backup in the Command pull-down menu	Select Backup
 Fig. 2:	Select backup device, e.g. CD-R the Drives pull-down menu	CD-Burner is associated with drive letter S
CustomerProtocols	Select Customer-Protocols in the Packages pull-down menu Click Go	Estimated back-up time is ~1 minute
	Wait for message: "End Backup"	
SW-Settings 02	Select SW-Settings 02 in the Packages pull-down menu Click Go	Estimated back-up time is ~4 minutes
	Wait for message: "End Backup"	
Numaris4	Select Numaris4 in the Packages pull-down menu Click Go	Estimated back-up time is ~20 minutes
	Wait for message: "End Backup"	

Output	Input	Note
Security Settings Only valid for systems with HIPAA	Select Security-Settings in the Packages pull-down menu Click Go	Estimated back-up time is ~1- 5 minutes
	Wait for message: "End Backup"	

NOTE

For more detailed instructions, refer to the manual “Technical Documents, Volume 1”, register “Installation Software”

3.2 Cleaning up directories

Required time: 10 min.

Required tools / parts: n.a.

SW Deleting the MR save log files



Fig. 5: Starting syngo MR Explorer

Starting the Explorer

- Press <Ctrl-Esc>
- Select <Start>, <Programs>, <Advanced User>, enter the password (see password list).
- Press <Ctrl-Esc>
- Select <Programs>, <Accessories>, <Windows Explorer>

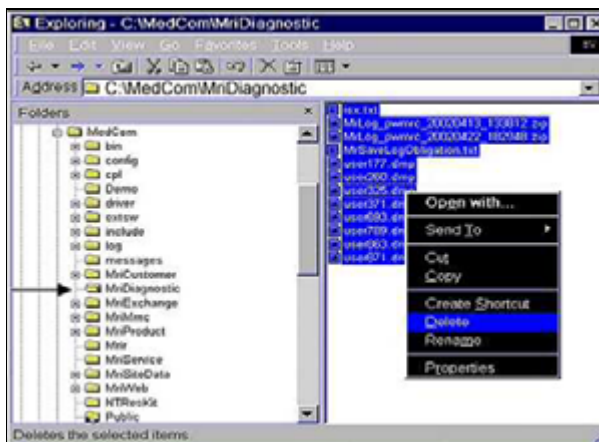


Fig. 6: Exploring C:\MedCom\MriDiagnostic

- Navigate to "C:\MedCom\MriDiagnostic"
- Click once in the right window and select all files <Ctrl-a>
- Right-click one of the files and select <Delete>.
- This opens the pop-up window "Confirm Multiple File Delete"
- Click <Yes> to delete the files
- Close the Windows Explorer

4 Image Quality Assurance

Required time: 70 - 120 min. depending on installed options

Required tools: Loaded body phantom, large spherical phantom with holder,

The quality assurance check has to be performed once a year but we **recommend** it be done every **three months**.

The quality measurement of the body coil is a fully automated procedure and the results are entered in the on-site database.

NOTE

Use the Online Help to obtain detailed information regarding, e.g. table control, phantom positioning, or specific items and procedures. Press the 'Help' button in the header bar of the corresponding platform.

Preliminary steps

Starting the service platform

All actions of the quality assurance procedure are performed via the service platform.

- Start the Service platform > **Options** > **Service** > **Local Service**.

From the Home menu, you access the **Quality Assurance** subplatforms by pressing the corresponding button.

Q

Quality Assurance Measurement

- Start the measurement by clicking the "Start" button.
- Follow the instructions provided by the system.
 - ⇒ The Quality Assurance check is successfully completed when **every** individual QA step has the status 'Done'.

NOTE

If the results are out of the tolerance, perform "Tune-Up" or troubleshooting as necessary. Perform the quality assurance check again and save the "site-specific data".

5 Changes to the previous version

No previous version exists.

